

CRITERION C SAMPLE:

Results:

Table 1. Bounce Heights of a Tennis ball before and after two hours of play

Bounce Height Before, Trial 1 (cm)	Bounce Height After, Trial 1 (cm)	Bounce Height Before, Trial 2 (cm)	Bounce Height After, Trial 2 (cm)	Bounce Height Before, Trial 2 (cm)	Bounce Height After, Trial 3 (cm)	Average Bounce Height Before (cm)	Average Bounce Height After (cm)
130	116	135	122	132	120	132	119

Summary of Trends in the Data:

Table 1 shows that the bounce height in all trials of the experiment was greater for the tennis ball before it underwent two hours of play time. In trials 1, 2, and 3 respectively, the tennis ball bounced 130cm, 135cm, and 132cm averaging 132 cm overall. After two hours of play Trials 1 to 3, showed results of 116cm, 122 cm, and 120 cm, averaging 119 cm overall. As shown in both Table 1 and Figure 1, the tennis ball bounced an average of 13 cm higher than the tennis ball after two hours of play.

Comparison, Analysis, and Conclusion:

The hypothesis for this experiment, which stated that **If** a tennis ball is dropped from the same height before and after two hours of play, **then** the unused tennis ball will bounce back higher **because** the rubber will lose elasticity after two hours of use was accepted based on the data collected.

Table 1 and Figure 1, as well as the summary of trends in the data, both show that the new tennis ball bounced higher than the used tennis ball in all three trials of the experiment. This makes sense since tennis balls are made of a special rubber compound with a hollow core filled with pressurized air. As the tennis ball is hit over and over, it will lose air as it deforms in shape from being hit. It will also suffer from elastic fatigue. This means that it won't be able to snap back into shape as easily after a lot of use (Mann). Due to these reasons, the bounce height will decrease over time, as shown in Table 1 and Figure 1.

Evaluation of the Method:

Error 1:

Source of Error	The Problem Caused by this error:	Way to fix or minimize the error
The way that the meter stick was attached to the wall (using double sided tape) and the one time use of the level	Tape can become loose and this may cause the meter stick to be uneven on the wall leading to misreading of the measurement. Although a level was used to ensure that the meter stick was level, it was only used at the initial attachment of the meter stick and not re-checked after each trial	• Use a more permanent adhesive to attach the meter stick to the wall (such as super glue) • Check the meter stick with the level after each trial

Error 2:

Source of Error	The Problem Caused by this error:	Way to fix or minimize the error
Human error - A human was used to drop the ball from the height of 1 m.	As people are not machines, it is impossible to ensure that the ball was dropped from the exact same height each time. This means a control variable may not actually be controlled and differences in drop height may have lead to data that is not as accurate as it could have been if a better method of dropping the ball had been used	• Use a machine to drop the ball that takes its own measurements (if available) • If you cannot get a machine, make multiple checks to ensure that the ball is dropped from the same height before dropping

MLA Works Cited

Admin. "The Science Behind Tennis Ball Bounce." *World Tennis*. World Tennis Magazine. 2025.
<https://worldtennismagazine.com/the-science-behind-tennis-ball-bounce/24358>

"Can you give me some reasons that I would want to know how much bounce a tennis ball will have before and after 2 hours of play. Prompt." Google Gemini. April 27, 2026. gemini.google.com

Mann, Alex. "The History of Tennis Balls." *Tennis Nerd*. Tennisnerd.net 2022.
<https://www.tennisnerd.net/balls/the-history-of-tennis-balls/28277>

"The Science of Tennis Balls." *Tennis Shop*. April 27, 2026.
<https://tennisshopblogs.wordpress.com/2024/03/08/the-science-of-tennis-balls/>